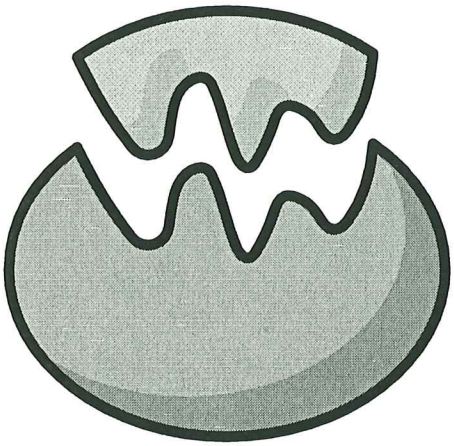
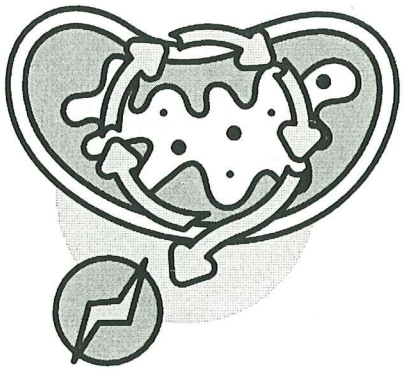
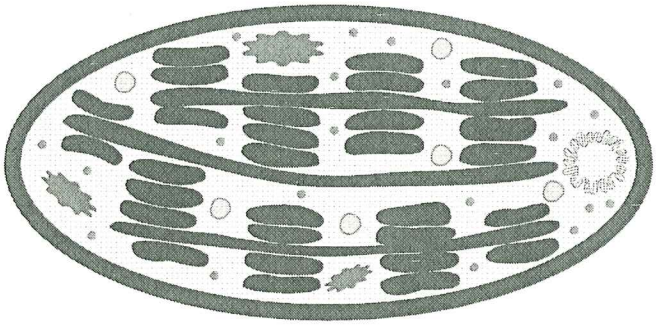
Unit 3 Lab Packet

Name:_____







Labs:

Enzyme Liver Lab

Poison Case Study

Photosynthesis Case Study

Algae Lab

Enzyme Investigation LAB

Introduction

What would happen to your cells if they made a poisonous chemical? You might think that they would die. In fact, your cells are always making poisonous chemicals. They do not die because your cells use enzymes to break down these toxins into harmless substances. Enzymes are proteins that speed up the rate of reactions that would otherwise happen more slowly. The enzyme is not altered by the reaction. You have hundreds of different enzymes in each of your cells. Each of these enzymes is responsible for a specific reaction that occurs in the cell.

In this lab, you will study an enzyme that is found in the cells of many living tissues – catalase. Catalase speeds up a reaction that breaks down hydrogen peroxide, a toxic chemical, into 2 harmless substances – water and oxygen gas. The reaction is as follows:

$$2\ H_2O_2 \rightarrow 2\ H_2O + O_2$$

This reaction is important to cells because hydrogen peroxide is produced as a by-product of many normal cellular reactions. If the cells did not break down the hydrogen peroxide, they would be poisoned and die. In this lab, you will study catalase found in liver cells. This enzyme is also found in potatoes.

READ ALL DIRECTIONS BEFORE DOING ANY EXPERIMENTS

Before doing the experiments below, label your test tubes A, B, C, D, and E

PART A – Observe normal Catalase reaction

1. Using forceps, put a piece of liver or potato into **test tube A**
2. Place 2 ml of the hydrogen peroxide solution into **test tube A**
3. Time the reaction for 60 seconds and then measure the height of foam in cm → Write this in **Table 1**.
4. Pour off the liquid from **test tube A** CAREFULLY into **test tube B**. DO NOT REMOVE THE LIVER or potato.
5. Pour 2 more mL of hydrogen peroxide onto the liver or potato in **test tube A**.
6. Time the reaction again for 60 seconds and then measure the height of foam in cm and write this in **Table 1**.
7. Put another piece of liver or potato into the liquid in **test tube B**
8. Time the reaction again for 60 seconds and then measure the height of foam in cm and write this in **Table 1**.

9. Answer the questions below:
    - What gas is being released through the bubbles?

- Looking at your reactions, is catalase reusable? Why or why not?

- Did the liver react with the used liquid in tube B? How do you know?

- Is substrate reusable? Why or why not?

PART B – What is the effect of temperature on Catalase activity?

1. Obtain a piece of boiled liver or potato and put it into **test tube C**
2. Pour 2 mL of hydrogen peroxide into **test tube C**
3. Time the reaction for 60 seconds and then measure the height of foam in cm and write this in **Table 1**.
4. Obtain a piece of frozen liver or potato and put it into **test tube D**
5. Pour 2 mL of hydrogen peroxide into **test tube D**
6. Time the reaction for 60 seconds and then measure the height of foam in cm and write this in **Table 1**.
7. Answer the questions below:
    - What did hot water do to the enzyme? How did this affect its activity?

    - Did freezing the liver affect its activity? Why or why not?

PART C – What is the effect of pH on Catalase activity?

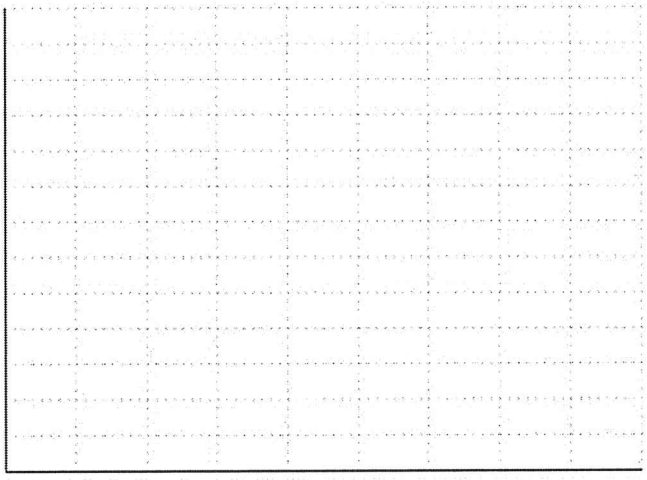
1. Add a piece of lemon soaked liver to **test tube E**
2. Pour 2 mL of hydrogen peroxide into **test tube E**
3. Time the reaction for 60 seconds and then measure the height of foam in cm and write this in **Table 1**.
4. Answer the questions below:
    - What was the effect of a low pH on catalase activity?

    - How does pH affect an enzyme?

Data Table

Table 1.		
	Sample	**Height of foam (cm)**
PART A	Test Tube A- Normal activity	
	Test Tube A- Reused Catalase	
	Test tube B-Reused Hydrogen peroxide	
PART B	Test Tube C- Boiled catalase	
	Test Tube D- Frozen Catalase	
PART C	Test Tube E- Catalase in an acid	

Graph your data below:

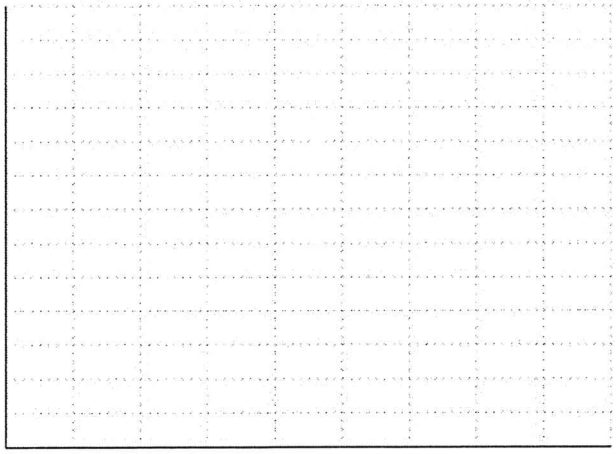


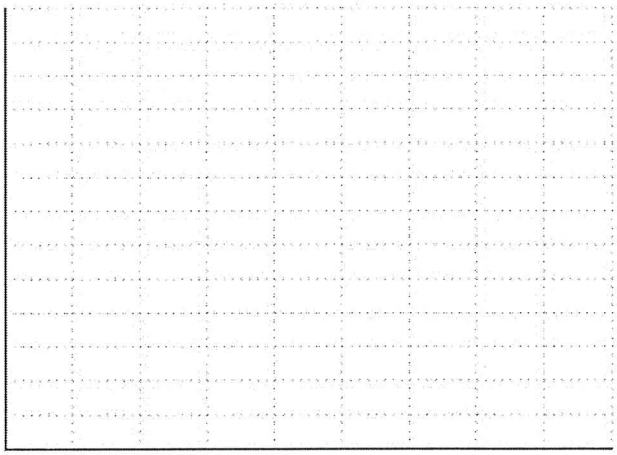
Part D: Interpreting Enzyme Data

The data table below shows the results of a study done on pH and temperature on catalase. Graph the data on the next page then answer the questions that follow.

pH	Rate of reaction
1	0
2	0
3	10
4	50
5	75
6	90
7	100
8	50
9	20
10	10
11	0
12	0

Temperature	Rate of Reaction
90	0
92	0
94	20
96	75
98	110
100	100
102	55
104	25
106	10
108	0





1. At what pH does catalase work best?

2. At what pH does catalase start to denature? How do you know?

3. At what temperature does catalase work best?

4. Your normal body temperature is about 98.6. When you are sick, your body raises your internal body temperature in order to help destroy pathogens. What would happen to catalase if your temperature spiked to 104?

5. Proteins make up the majority of structures in your body. Using what you learned about enzyme shape, why would a high fever negatively impact your body and these proteins?

Name:_____ Period_____

ATP CANDY MODELING

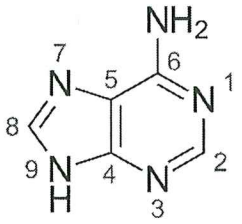
Essential Question: What is ATP and how is it used by living cells?

ATP, or adenosine triphosphate, is the main energy-carrying or energy-transferring molecule of all living things. In cellular respiration, the energy in the food you eat is released and used to power cellular activity. ATP is the molecule that accepts energy from the food when it is broken down and transfer it to where it is needed in the cell. ATP is considered the energy "currency" of the cell that is "spent" when a cell needs to do work.

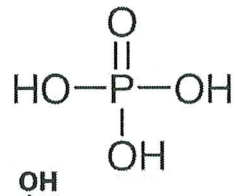
ATP has three subunits: an adenine, a ribose sugar, and 3 phosphate groups. "Adenosine" is the term used for the molecule formed when adenine is bonded to ribose.

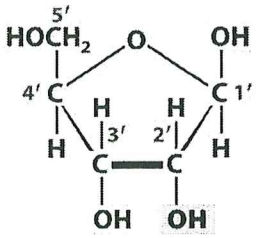
Part A: Chemical structure of ATP

Directions: Look at the below to the right and answer the following questions.



1. What is the molecular formula of ribose? C___H___O___
2. What is the molecular formula of adenine C___H___N___
3. What atom is present in adenine that is not present in ribose?
4. What atom is present in ribose that is not present in adenine?
5. What is the molecular formula for a phosphoric acid? H___P___O___





Ribose

Part B: Constructing an ATP Molecule

Directions: Follow the steps below to construct a model of ATP. Answer the questions that follow.

1. Obtain the following: 1 small plate, 3 mini marshmallows, 1 large marshmallow, 1 gummy bear, and 2 toothpicks
2. Take the big marshmallow and stick a toothpick in each side of the large marshmallow
3. On the left side, attach the gummy bear to the toothpick
4. On the right side attach 3 mini marshmallows.
5. Draw and label your ATP model below:

6. What does each of the following represent in your model:
   a. The big marshmallow:
   b. The 3 small marshmallows:
   c. The gummy bear:
   d. The toothpick:

Part C: Using ATP to Power Reactions

The usable energy in ATP is stored in the last phosphate off an ATP molecule and the energy is released. The molecule that is leftover is called Adenosine Diphosphate since it only has two phosphate groups left.

1. Pull the last mini marshmallow off of your ATP model and set it to the side
2. Draw and label what your model looks like now below:

3. Fill in the chemical equation below:

   ATP → _____ + _____ +_____

4. What is the reactant in the equation above?
5. What are the products?
6. Which product will be used by the cell?

Part D: Reusing ADP

ADP does not have enough energy to power the cell, but we can "reenergize" it by adding another phosphate group to it. The energy needed to do this comes from two places. In animals, we breakdown the food we eat through cellular respiration and that helps us to turn ADP to ATP. Plants also use cellular respiration, but they can reenergize a small amount of ATP by also using photosynthesis.

Its estimated that humans used almost their body weight in ATP each day to keep their cells functioning. In fact, a cell can use about 2 trillion molecules of ATP per second!!! Where does all of this ATP come from? From recycling ADP over and over again, kind of like recharging a battery.

Directions: Follow the steps below then answer the questions that follow.

1. Eat the mini marshmallow "phosphate group" that you removed. This will give you the "energy" to recharge ADP.
2. Add another mini marshmallow to the end of the toothpick on the right side. You now have a "recharged" ATP!
3. Fill in the chemical equation below:

   _____+ _____ + _____ →

4. In the space below, draw the ADP/ATP cycle. Show where energy is leaving and entering the cycle.



Poison Case Study



Essential Question: How would a disruption to cellular respiration affect organisms?

Adapted From: Cellular Respiration Gizmos

Background: Cellular respiration is the process that all eukaryotes use to turn energy from food into chemical energy. These reactions take place in the mitochondria and produce ATP for cells and tissues. Mitochondria are especially important in animals in muscular and cardiac tissue. For example, your heart requires a massive amount of ATP because its cells are constantly keeping your heart beating. Any disruption to this process is eventually going to lead to not enough ATP being produced. This can lead to severe disease and possibly death.

There are many things that can impair the function of the mitochondria including cancer, genetic disorders, and even certain types of poisons. Different poisons affect the mitochondria in different ways. When someone has accidentally ingested something poisonous, a doctor called a Medical Toxicologist has to figure out what part of the mitochondria is being affected through medical tests.

Directions: You will be assigned a case study on a person or group of persons who have suffered from poisoning. The victims in each case study died, but their deaths are unresolved. Your job is to find out why they died and what type of poison they might have ingested.

You will be learning about the different steps to cell respiration so that you can figure out what part of your victims' mitochondria has been compromised. This data collection will take place over several class periods. You will complete each part of this handout as we explore each step to cellular respiration.

Materials: One envelope with the case study assignment, background information on poisons, and data sets.

Case Study Assignment #_____

In the table below, summarize your case study:

Victim(s)	
Symptoms	
Background Info	

Part A: Cellular Respiration Overview

1. Define cellular respiration:

2. Write the equation (in words AND in chemical formula below)

3. Two reactants are needed for cellular respiration. State these two reactants and their source (where they come from):

   - Reactant 1:
   - Reactant 2:

4. What are the products of cellular respiration and where do you think they go?

   - Product 1:
   - Product 2:

5. What is also produced that is NOT part of the equation above?

Part B: Summarizing the Steps to Cellular Respiration

As you learn about each step of cellular respiration fill in the table below. This will take several class periods to complete.

Step 1: Glycolysis

Location	
Input(s)	
Output(s)	
Predict: If cellular respiration was interrupted at this step, how would that affect the products of glycolysis?	

Step 2: Pyruvate Oxidation

Location	
Input(s)	
Output(s)	
Predict: If cellular respiration was interrupted at this step, how would that affect the products of pyruvate oxidation?	

Step 3: Krebs Cycle

Location	
Input(s)	
Output(s)	
Predict: If cellular respiration was interrupted at this step, how would that affect the products of the Krebs cycle?	

Step 4: Electron Transport Chain

Location	
Input(s)	
Output(s)	
Predict: If cellular respiration was interrupted at this step, how would that affect the products of the ETC?	

Part C: Evaluating Poisons

Using the information on poisons given to you by your instructor, complete the following table.

Poison	**Mechanism of Action (what step does it interrupt)**	**Effect on cellular respiration**
2-deoxyglucose		
Arsenic		
Cyanide		
Oligomycin		

Part D: Evaluating Data

You will be evaluating toxicology results for your victim(s) and comparing them to that of a control group. You will be given a sample from healthy muscle cells.

Record the **data** for the healthy cells below. Next compare the data to the data on poisons and put a downward arrow (↓) if that molecule is decreasing, an upward arrow (↑) if it's increasing, or an equal sign (=) if it stays the same.

Table 1: Predictions

Product	**Healthy Muscle Cells**	Cells affected by **2-Deoxyglucose**	Cells affected by **Arsenic**	Cells Affected by **Cyanide**	Cells Affected by **Oligomycin**
Pyruvate					
NADH					
Intermembrane H+					

Next chart the victims' data and state how it is <u>different</u> from the data above (Is it the same, higher, or lower)?

Table 2: Victim Evaluation

Product	Victim's Muscle Cells	How is it different from healthy muscle cells?
Pyruvate		
NADH		
Intermembrane H+		

<u>Part E: Conclusions</u>

1. What poison is affecting your victim(s)? **EXPLAIN HOW** you know it is that poison.

2. Put yourself in the shoes of a medical toxicologist. Write out the explanation you would give to the family of the victim(s) on how they died.

3. Why was it important to compare your victim's data to the healthy cell data (the control group)?

4. What is the importance of the energy carriers (NADH, $FADH_2$)?

5. What is a proton gradient and why is it important to cellular respiration?

6. Why isn't glycolysis enough to support human cells? Why do we need the other steps?

Photosynthesis STEM Case

Go to explorelearning.com and login. Start with the "Photosynthesis STEM Case" which will prompt you to the "Handbook". Once you've completed all the Handbook section, you do **NOT** have to continue with the Photosynthesis STEM Case experiment. Answer the questions below as you go.

All work must be included in your lab notebook and all questions answered in complete sentences. Use the same headings and question numbers as below:

History

1. Are corals considered animals or plants?

2. Review: What type of **symbiosis** describes the relationship between algae and coral? Explain what each organism provides to the other.

Background

3. What is the problem you are investigating?

"Handbook" – The Basics

4. Write the equation for photosynthesis. Be sure to have the reactants (inputs) on the left and products (outputs) on the right of your equation.

5. What is the relationship between photosynthesis and cell respiration?

"Handbook" – Inside the Chloroplast

6. Draw a sketch or find a picture of a chloroplast and label a **granum** and a **thylakoid.**

"Handbook" – Using Light Energy

7. Why does chlorophyll appear green? What is the role of chlorophyll?

8. Why does Photosystem II come before Photosystem I?

"Handbook" – Photosystem II

9. What type of molecule is ATP Synthase? What is its role?

10. As electrons are moved through the electron transport chain, where is the higher concentration of H+, inside or outside of the thylakoid?

"Handbook" – Photosystem I

11. Which of the following is *false* about PSI?
    a. PSI splits water to get electrons.
    b. PSI contains the pigment chlorophyll.
    c. PSI is needed to make NADPH.

d. PSI uses light energy to energize electrons.

"Handbook" – The Calvin Cycle

12. How many carbon atoms are found in 1 molecule of glucose? _____

13. Where does carbon dioxide enter in a plant?

14. What is Rubisco? What is its role?

15. Which of the following is true about the Calvin Cycle?
    a. Rubisco fixes oxygen to a carbohydrate molecule.
    b. Both ATP and NADPH are needed to make sugars.
    c. Light energy is converted to chemical energy.

Complete the Review to check your understanding.

1. What would happen to the concentrations of ATP, NADPH, and Sugars if PSII stopped working?
    a. ATP - _____
    b. NADPH - _____
    c. Sugars - _____

2. What would happen to the concentrations of ATP, NADPH, and Sugars if PSI stopped working?
    a. ATP - _____
    b. NADPH - _____
    c. Sugars - _____

3. What would happen to the concentrations of ATP, NADPH, and Sugars if the Calvin Cycle stopped working?
    a. ATP - _____
    b. NADPH - _____
    c. Sugars - _____

Name: _____ Date:_____

Investigation: Algae Bead Photosynthesis & Respiration

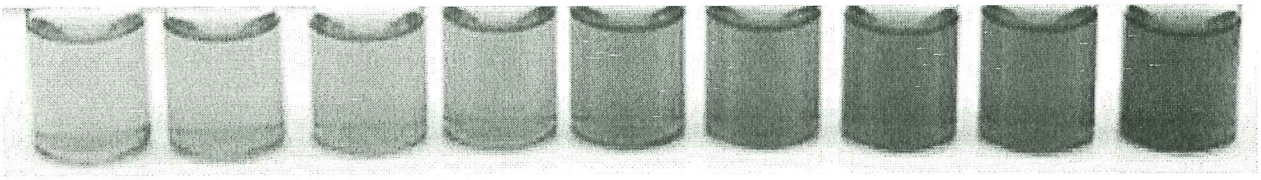
Heterotrophs (animals) are dependent on photosynthesis carried out by autotrophs (plants) for their energy needs. The overall equation for photosynthesis is written as:

$$6\,CO_2 + 6\,H_2O + \text{light energy} \longrightarrow C_6H_{12}O_6 + 6\,O_2$$

In this lab, you will use algae beads to indirectly measure the rates of photosynthesis by monitoring the amount of carbon dioxide produced. The algal beads will be placed in a carbonate indicator solution.

This solution should change color during some stages of your investigation because it is sensitive to pH changes caused by CO_2. When it is red it shows that it contains a concentration of CO_2 that is equivalent to the concentration of CO_2 in the atmosphere (0.04%). When there is more CO_2 in the indicator, it will turn from red through orange to yellow. When CO_2 is removed from the indicator solution, it will turn from red through magenta to dark purple.

Photosynthesis uses CO_2 (it should therefore be removed from the indicator solution by photosynthesising algae, causing the indicator to turn purple). Respiration will also be occurring in the algae which will release CO_2 (it should therefore be added to the indicator solution by respiring algae causing it to turn more yellow). In bright light the rate of photosynthesis in your algae should be greater than the rate of respiration, so a net reduction of CO_2 should occur.



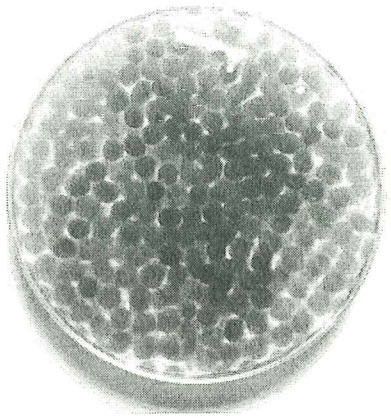
yellow		orange		red		magenta		purple
pH 7.6	pH 7.8	pH 8.0	pH 8.2	pH 8.4	pH 8.6	pH 8.8	pH 9.0	pH 9.2

Pre-Lab Questions:

1. How will the pH of the CO_2 indicator solution change as a result of photosynthesis?

2. If no photosynthesis is occurring, but the algae is still respiring, what color would you expect the indicator to change?

3. Under what conditions would you expect photosynthesis to occur?



4. Under what conditions would you expect respiration to occur?

   How will you know which is occurring in the tubes?

5. **Predict** what will happen to a tube that is placed in darkness compared to one that is placed under a light?

How Does Light Intensity Affect Photosynthesis?

Complete an experiment to answer the experimental question using the following materials:



- Algae Beads 5-8 per vial
- 2 ml of indicator solution per vile
- Light Source
- Aluminum Foil
- Distilled water to fill vial

Experimental Methods:

1. Grab 2 vials for your beads
2. Place 5-8 beads in each vial using the blue scoopula
3. Place 1 ml of bicarbonate indicator solution in the vial
4. Fill the rest of the vial with water all the way to the top.
5. Snap the vial shut tightly.
6. Cover one vial with aluminum foil, or put it in a dark place.
7. Leave one vial without any covering.
8. Record your initial observations in the chart below, use the indicator chart to measure the approximate pH.
9. Let the vials sit undisturbed until your next class
10. NEXT CLASS→BEFORE moving the vial notice any color change that occurred, and use the indicator chart to measure the approximate pH.

Write a **CLAIM** that answers the experimental question:

Gather **EVIDENCE** (observations) from the experiment. Organize your data onto a chart, using the pH scale given for color changes in the vials	Explain your **REASONING** for how the evidence supports your claim. Provide a **scientific explanation** for the results of your experiment.